

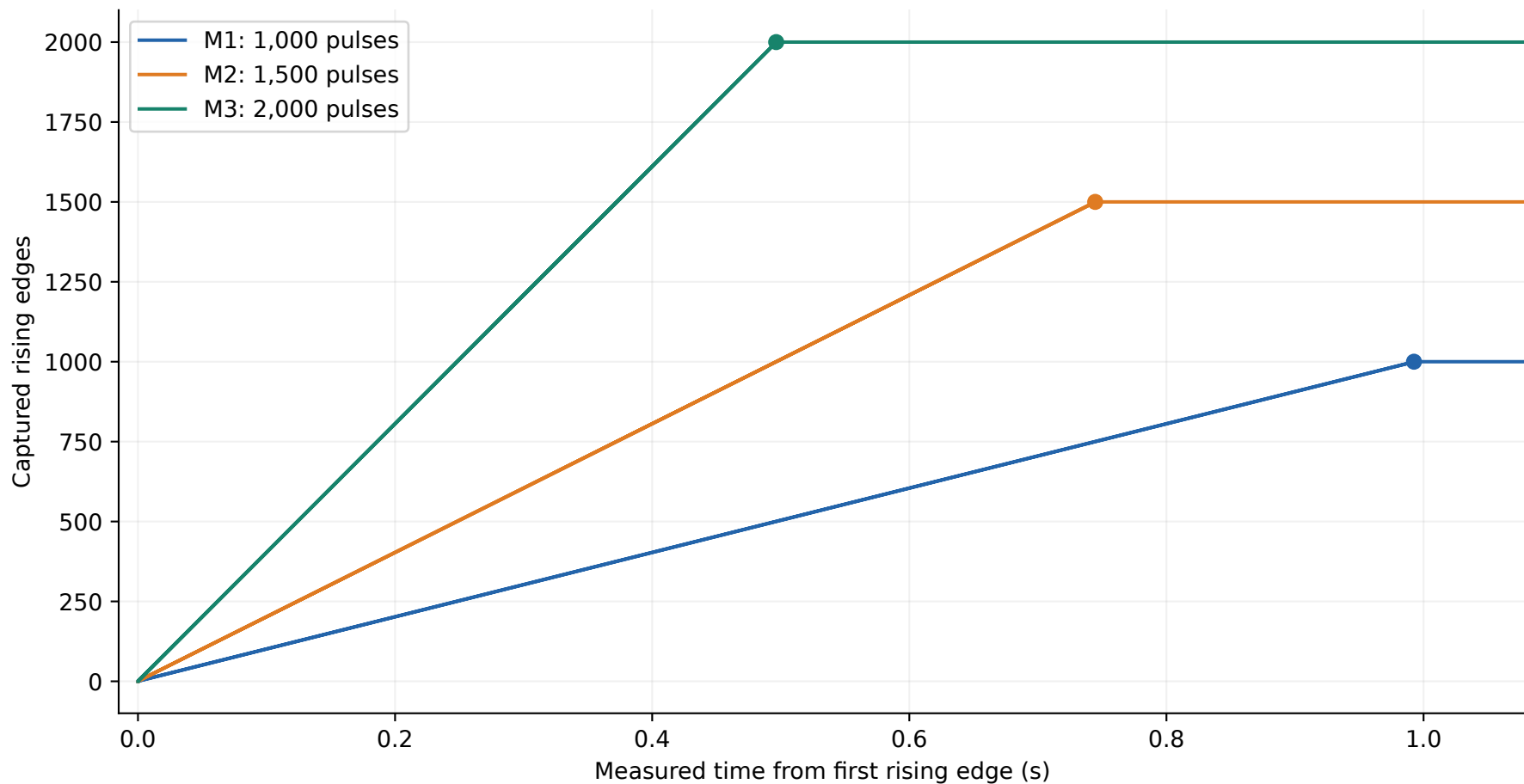
# STEP hardware validation — pulse counts and stopping

Current firmware: reported pulse counts match the independent logic-analyzer captures.

Absolute clock timing remains unresolved.

| Case | Captured M1/M2/M3    | Reported M1/M2/M3    | State     | Counts |
|------|----------------------|----------------------|-----------|--------|
| 1    | 1/2/3                | 1/2/3                | Completed | MATCH  |
| 2    | 1000/1500/2000       | 1000/1500/2000       | Completed | MATCH  |
| 3    | 100000/100000/100000 | 100000/100000/100000 | Completed | MATCH  |
| 4    | 1024/1024/1024       | 1024/1024/1024       | Completed | MATCH  |
| 5    | 995/498/249          | 995/498/249          | Stopped   | MATCH  |
| 6    | 1000/1000/1000       | 1000/1000/1000       | Completed | MATCH  |
| 7    | 512/256/128          | 512/256/128          | Underrun  | MATCH  |
| 8    | 435/435/435          | 435/435/435          | Completed | MATCH  |

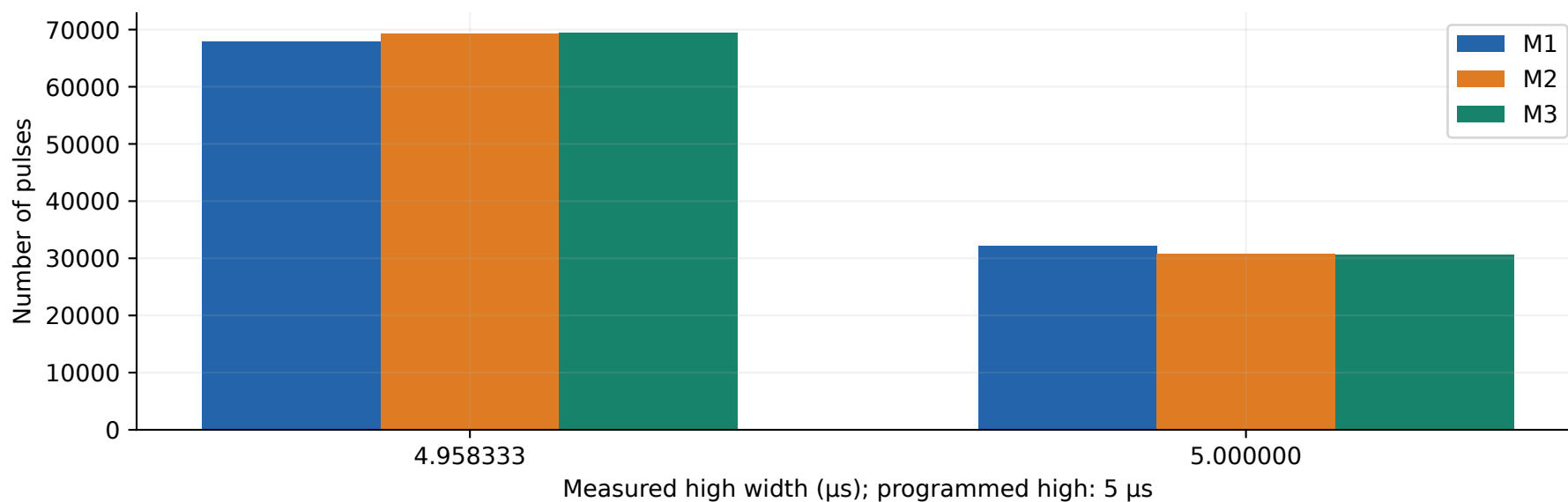
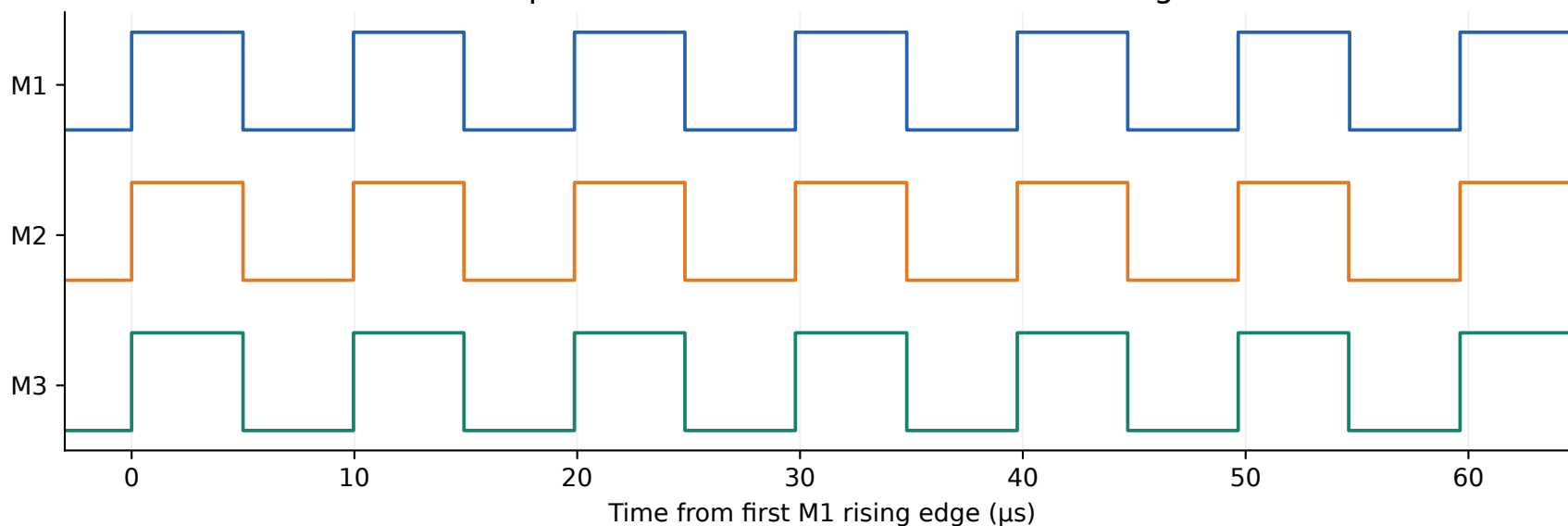
## Case 2 — independent rates and finish times



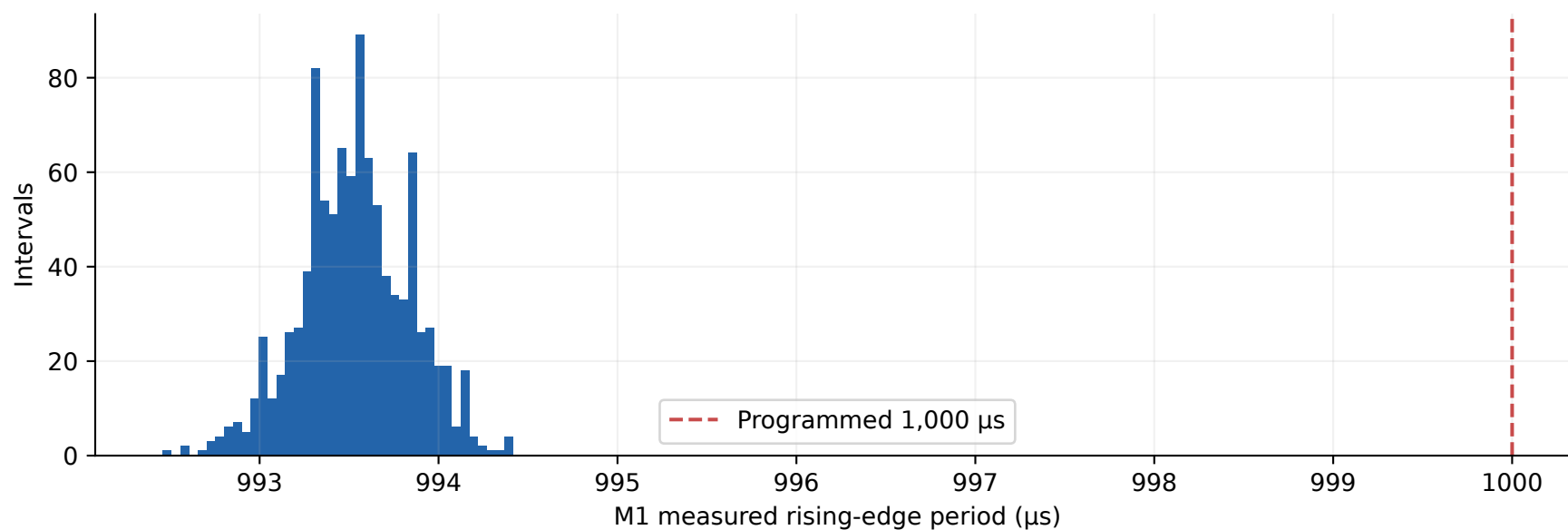
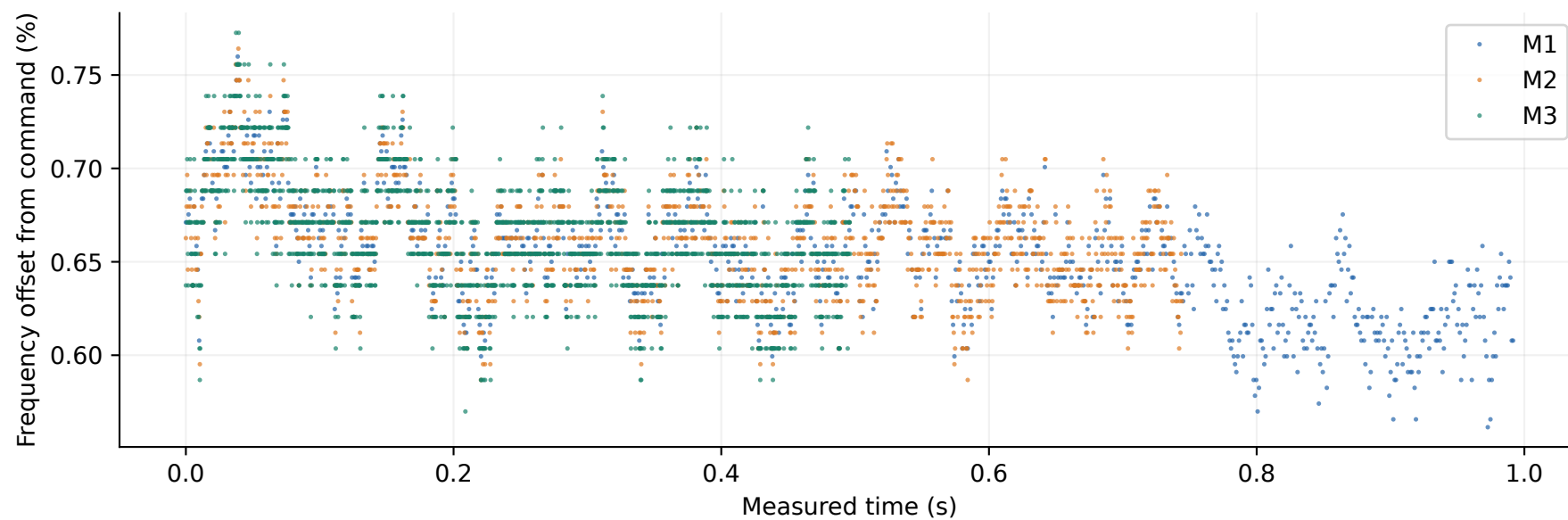
All first edges align within one 41.7 ns sample. Dots mark the final falling edges; no subsequent pulses in the capture.

# Case 3 — 100,000 pulses/axis; measured M1 rate 100.643 kHz

First pulses: all three axes share the first edge

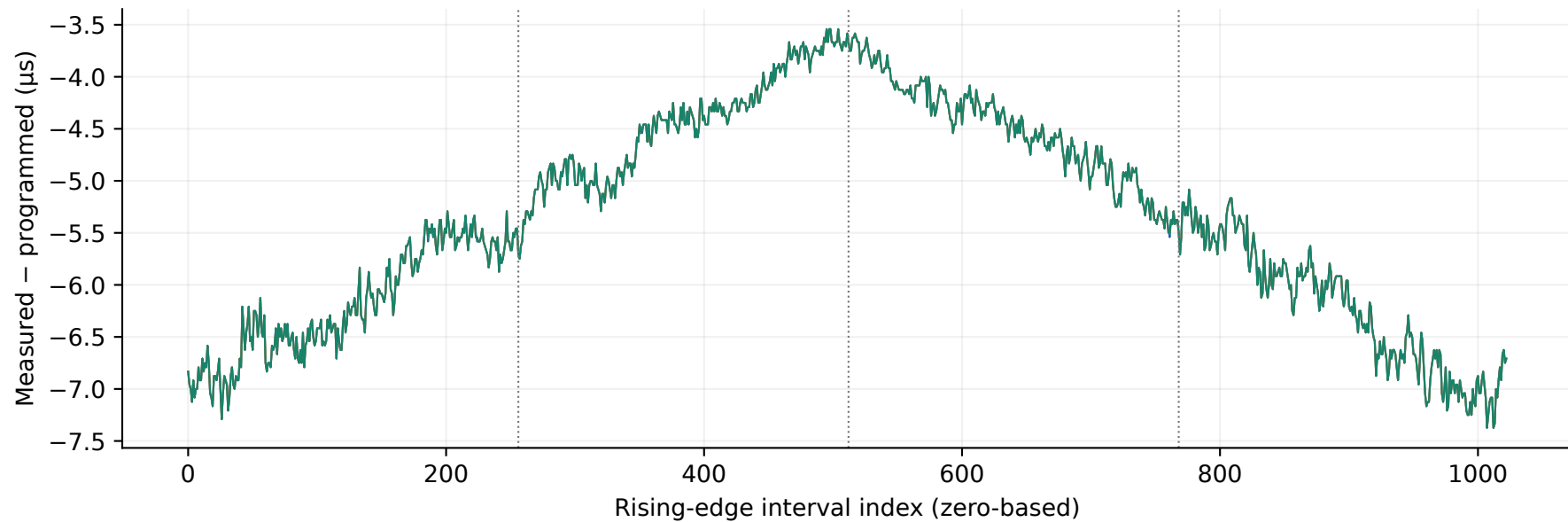
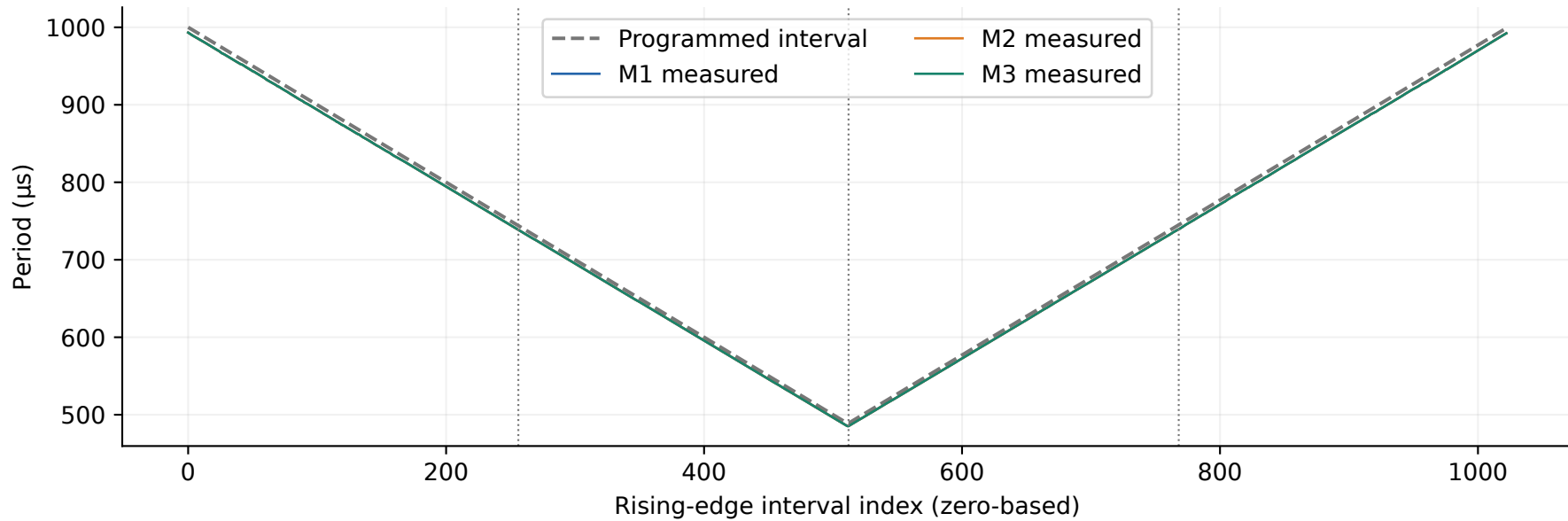


# Clock accuracy finding — frequency offset and variation remain



Timing is relative to the analyzer clock. Independent Hantek measurements also show a fast STEP rate; absolute calibration remains open.

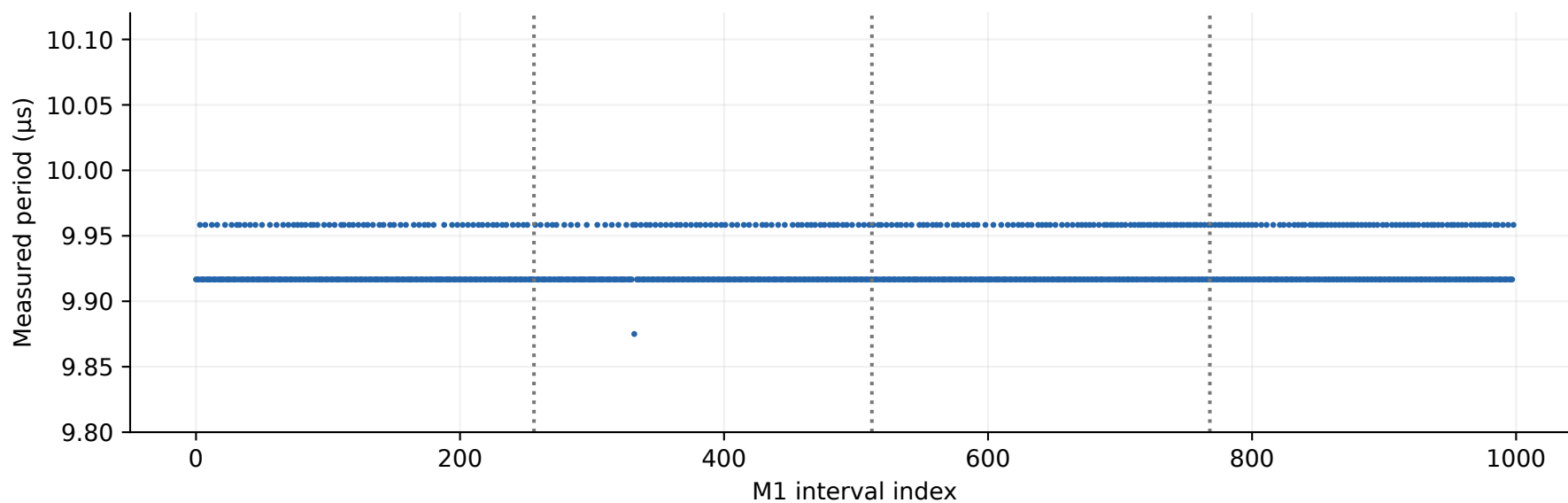
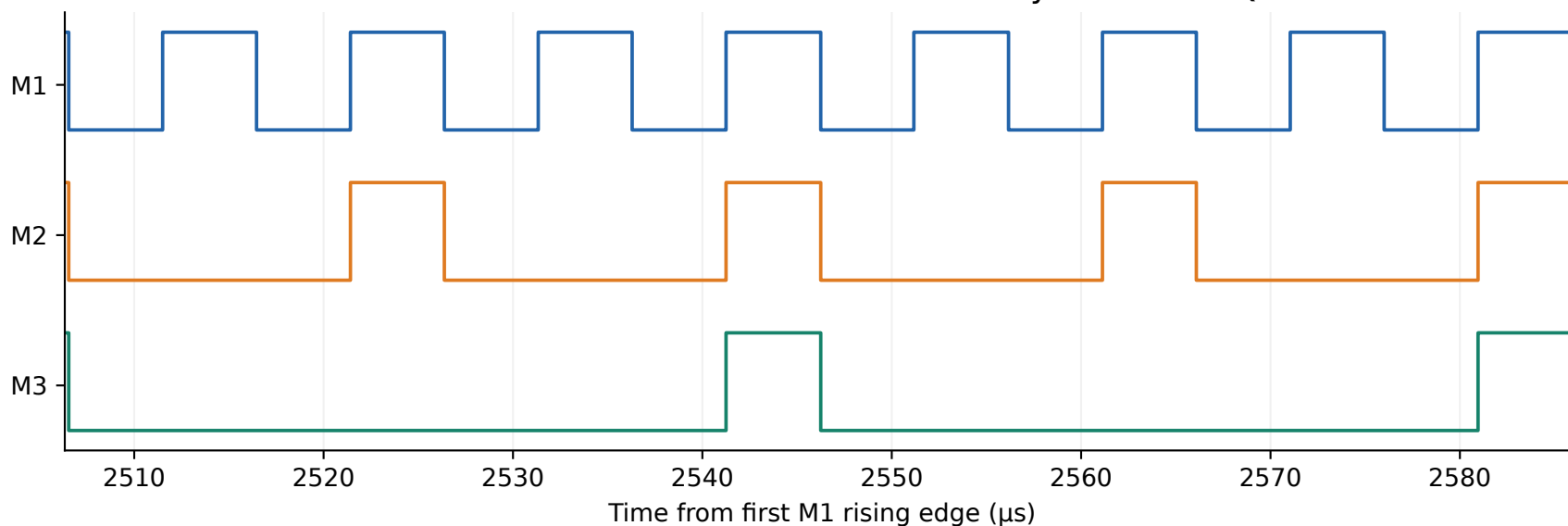
## Case 4 — copied acceleration/deceleration profile, 1,024 pulses/axis



Dotted lines indicate 256-pulse buffer boundaries. All counts match; systematic timing error remains visible.

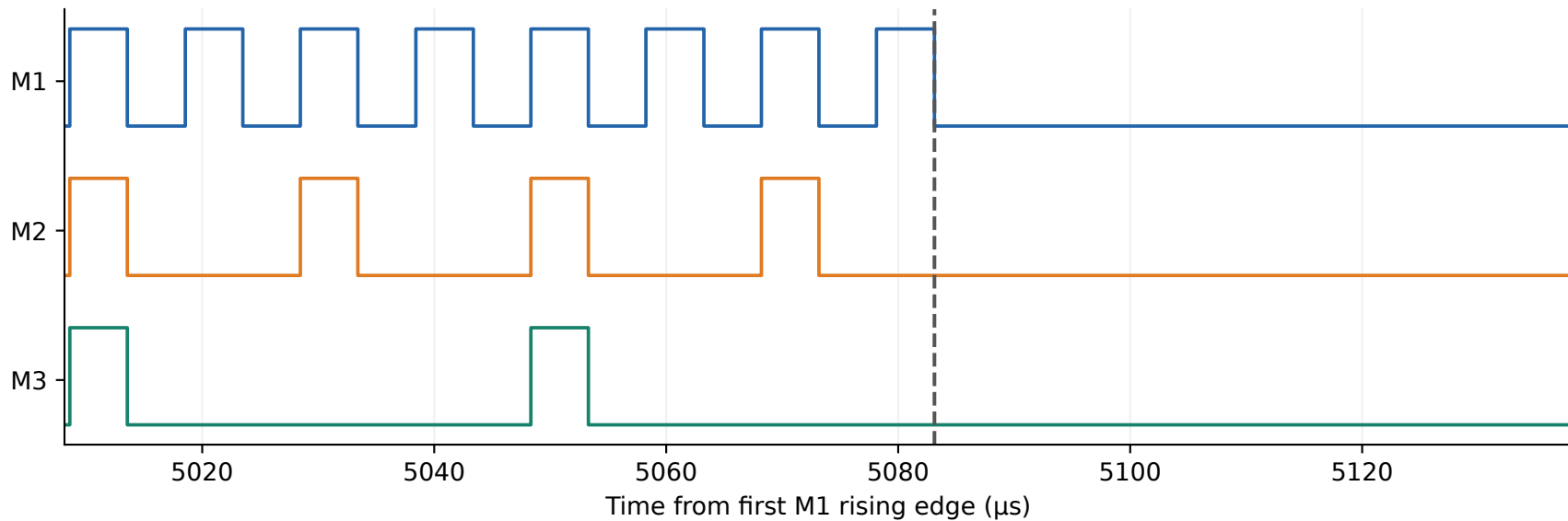
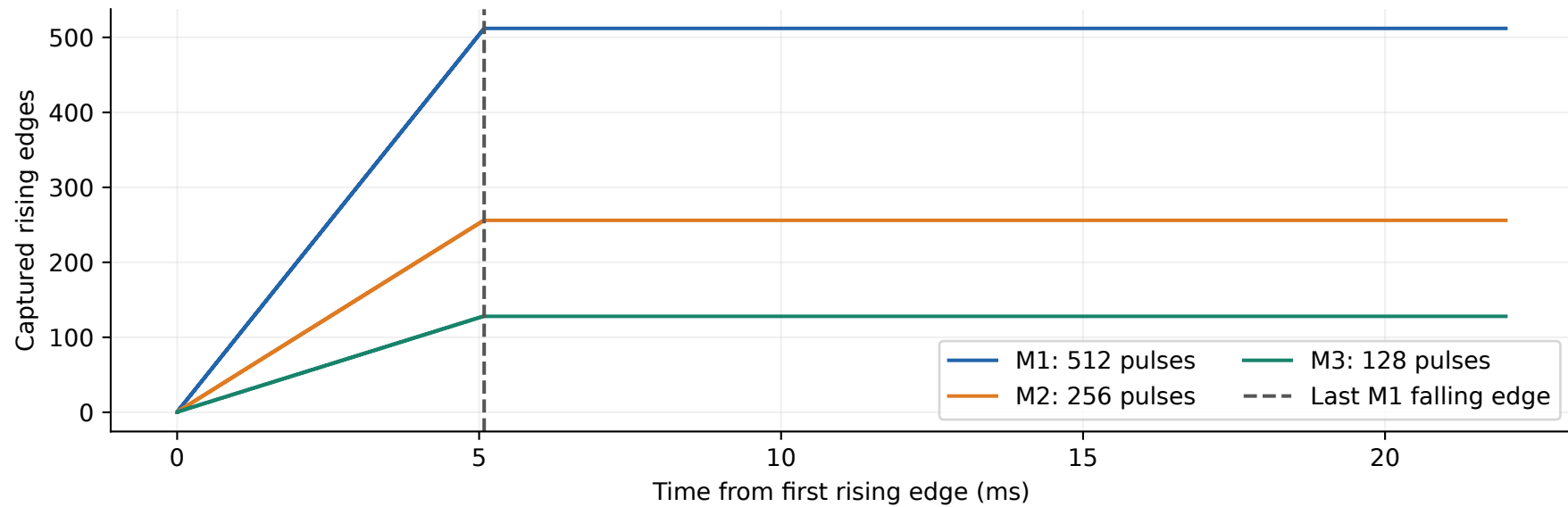
## Case 6 — delayed refill IRQ: 1,000 captured pulses on every axis

Pulse train crosses the first M1 DMA-buffer boundary while its IRQ is withheld



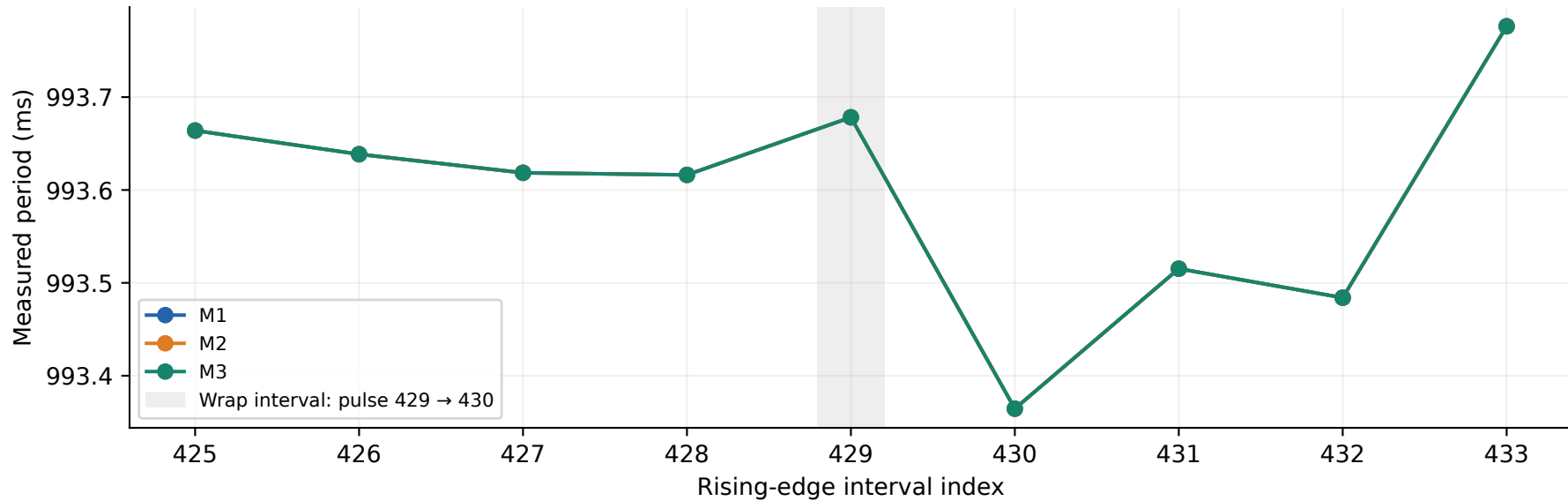
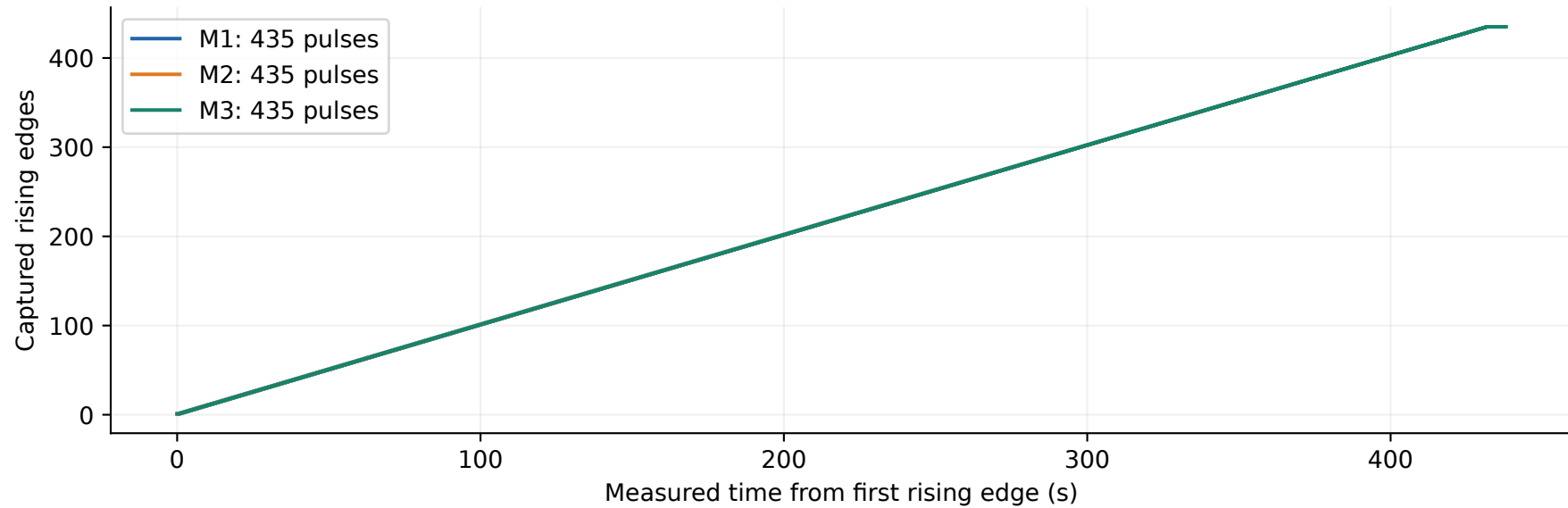
The firmware masks DMA IRQs for about 4.5 ms including the start delay. The analyzer observes the outputs, not the IRQ mask directly.

## Case 7 — autonomous stop with all DMA IRQs withheld



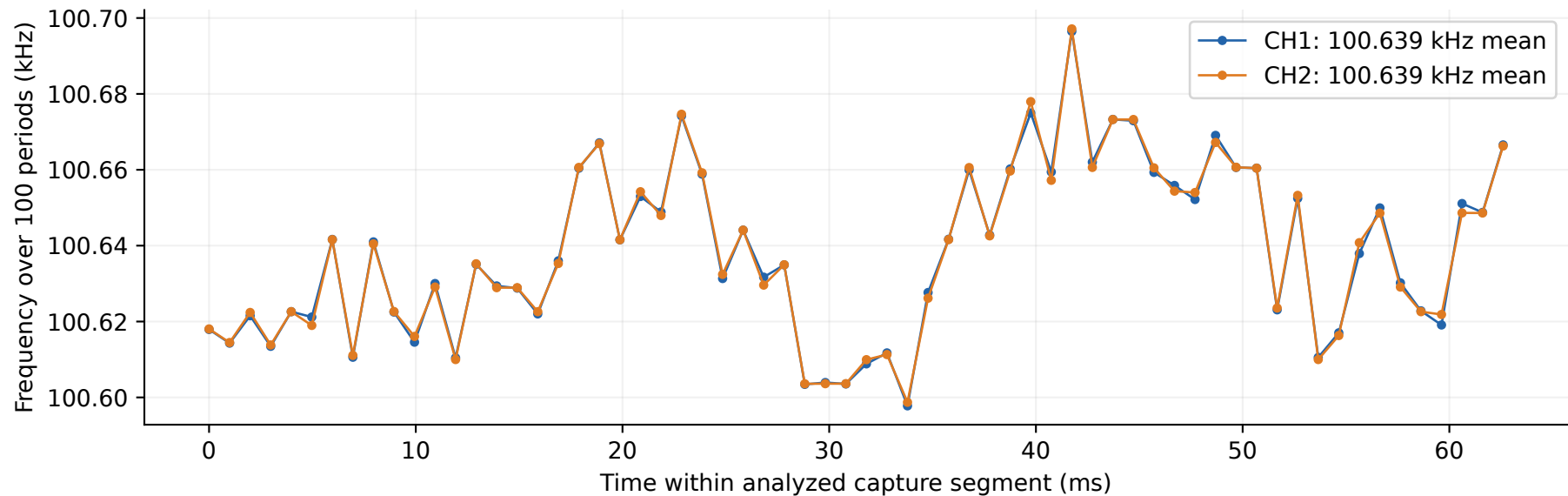
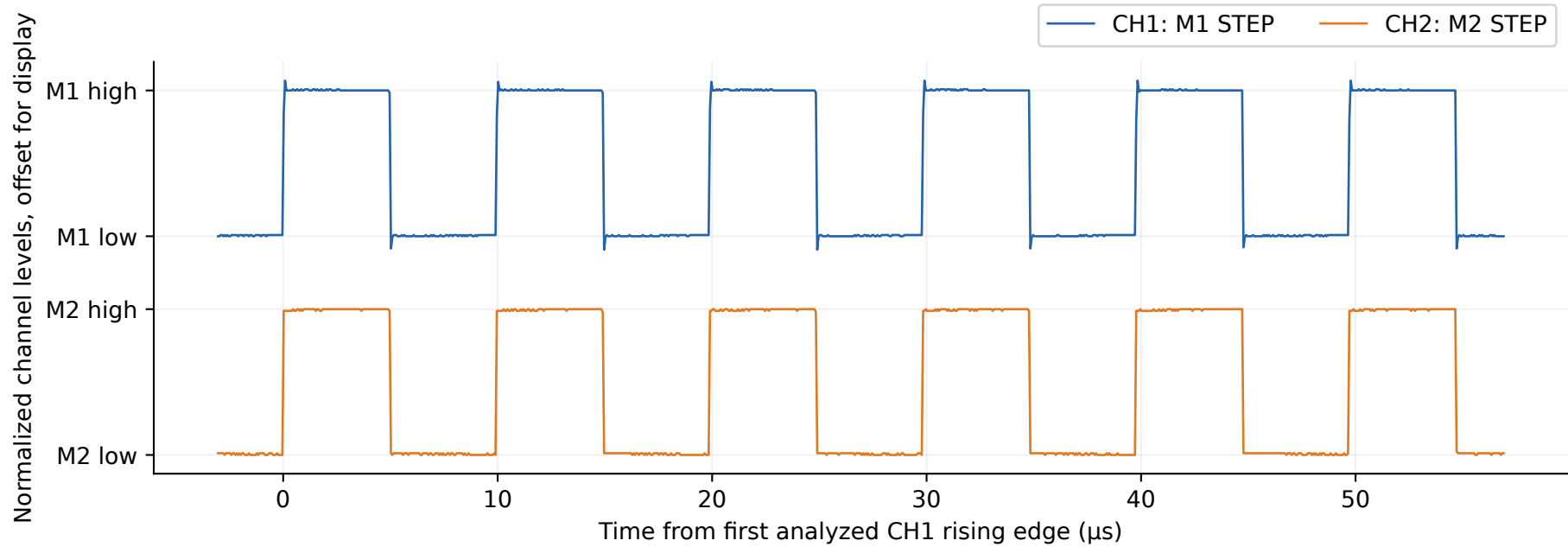
UART: Underrun, exact=1, counts=512/256/128, guard\_stopped=1 before IRQ restoration. Outputs remain low after the final pulses.

## Case 8 — real 32-bit timer rollover, 435 pulses on every axis

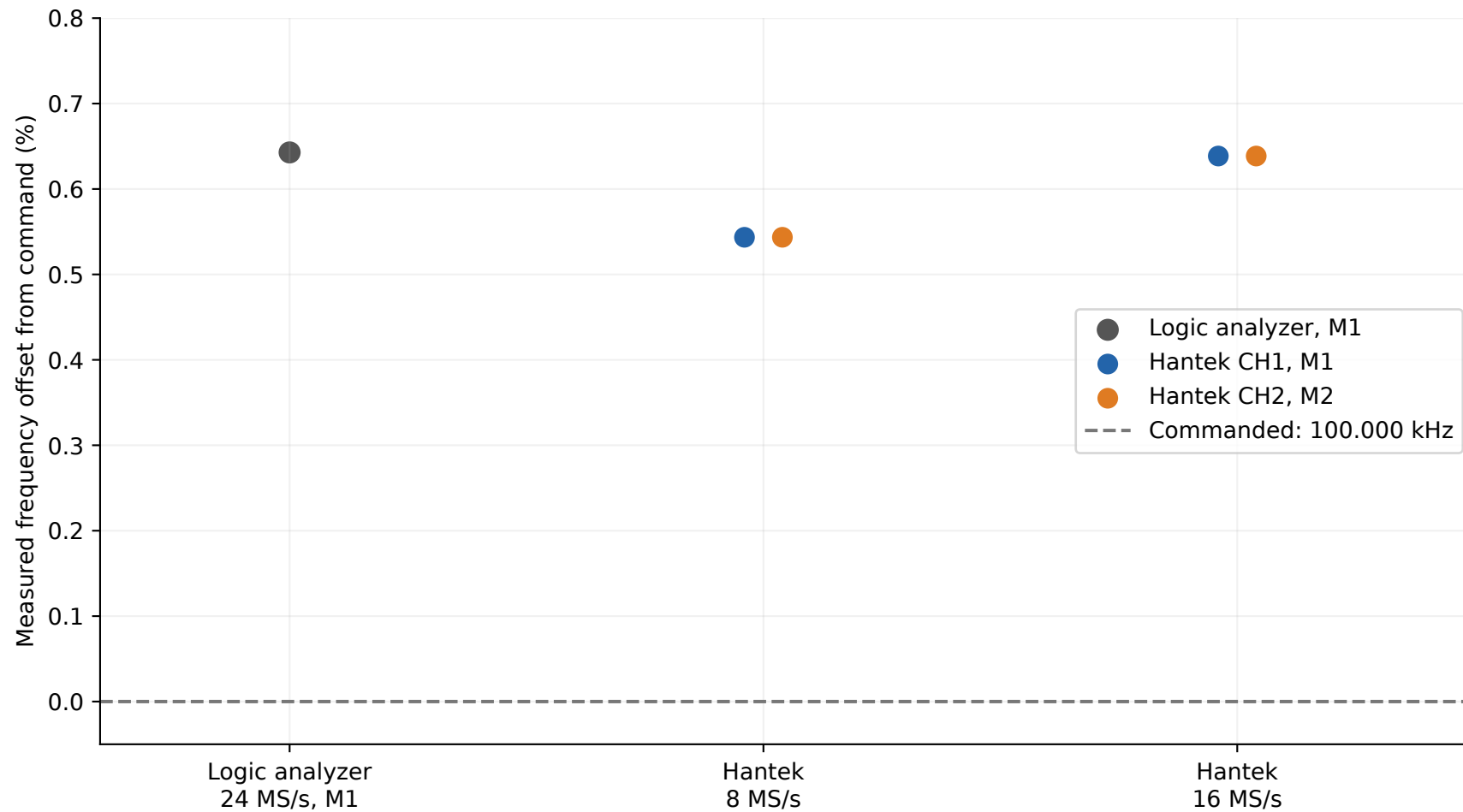


4 MS/s capture; 250 ns sample spacing resolves the 5  $\mu$ s highs. The rollover interval is located from scheduled timer ticks; no forced wrap.

# Hantek 6022BE — simultaneous M1 and M2 STEP measurements



# Independent instruments both measure a fast STEP rate



Same firmware, separate bursts. Channel pairs are simultaneous within each Hantek capture. No instrument has an independent timebase calibration.